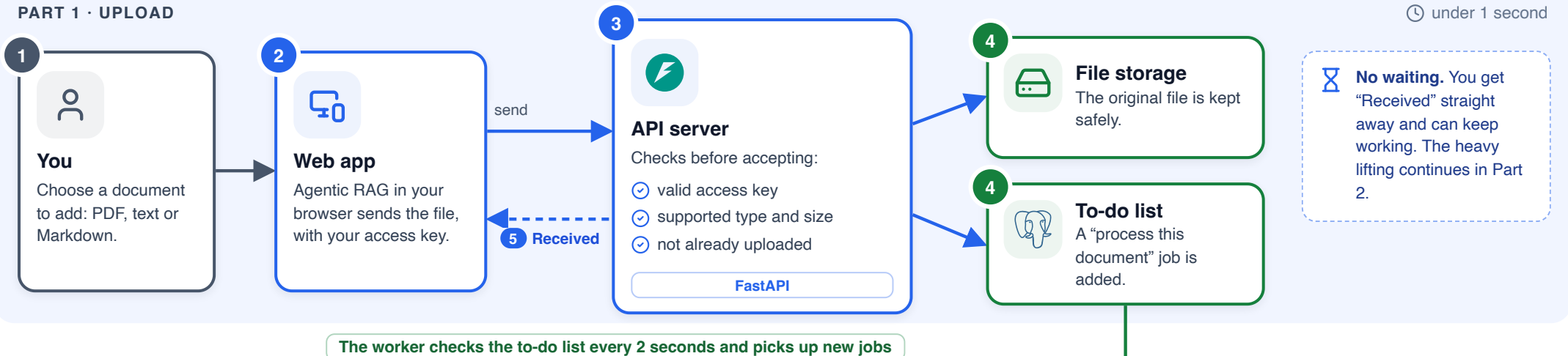


How a document becomes searchable

What happens, step by step, when someone adds a document to **Agentic RAG**, from the browser to the database.

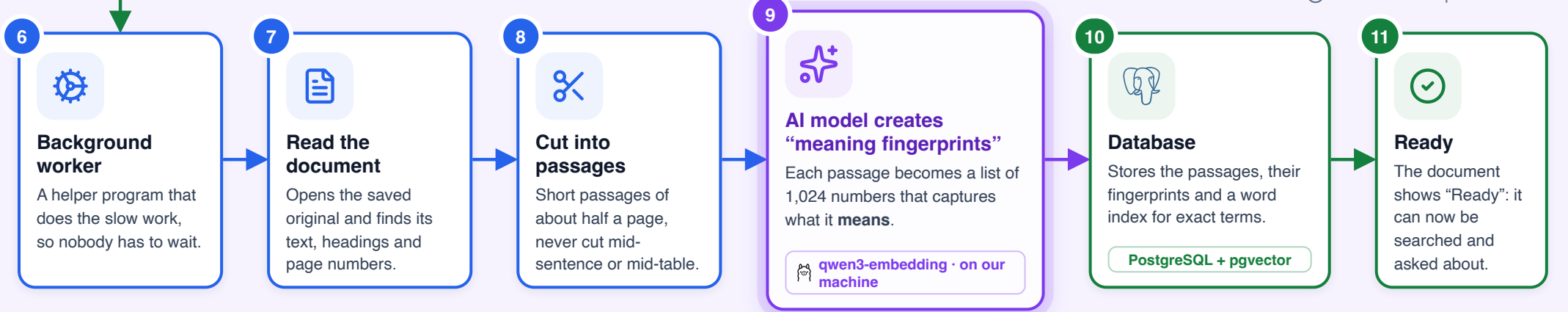
PART 1 · UPLOAD

🕒 under 1 second



PART 2 · PROCESSING IN THE BACKGROUND

🕒 a few seconds per document



Why is AI used here, and only here?

Fingerprints let the system find passages **by meaning, not just matching words**. Later, a question like “how do I stop browsers using old JavaScript?” finds the section on “cache busting”, even though they share almost no words. This is the **only AI step during upload**. The chat AI that writes answers is not involved until someone asks a question.



The same journey, in plain language

Example from our sample collection: 4 documents became 32 passages, at about 2.5 seconds per document.

STEP	WHAT HAPPENS	WHY IT MATTERS	TIME
1–2	You choose a document and the web app sends it.	Any PDF, text or Markdown file up to 25 MB.	instant
3	The API server checks it.	Only people with a valid access key can add documents; broken files and duplicates are rejected early.	instant
4	The original is saved and a to-do job is created.	Nothing is lost if something fails later, because the original is kept.	instant
5	You get “Received”.	You never wait for the slow part.	under 1 s
6	A background worker picks up the job.	Slow work happens separately, so the app stays fast for everyone.	up to 2 s
7	The document is read: text, headings, pages.	Headings tell us which section each passage belongs to; page numbers let answers cite “p. 12”.	under 1 s
8	It is cut into short passages.	Questions are answered from a few relevant passages, not whole documents.	under 1 s
9	AI turns each passage into a meaning fingerprint.	This is what makes search work by meaning instead of exact wording.	~2–3 s for a typical document
10	Passages, fingerprints and a word index are saved.	Two ways to search later: by meaning (fingerprints) and by exact terms (word index).	under 1 s
11	The document is marked “Ready”.	It can now be searched and asked about.	instant

✦ Where AI is involved

One AI model, at one step (9). It works like a librarian who files every page under the *ideas* it contains, not just the words it uses. That's why a question can find the right passage even when it is phrased differently.

The chat AI that writes answers, with references to these passages, is only used later, when someone asks a question.

✓ Safe and reliable by design

- Access keys are required; only a scrambled form of each key is stored.
- Files are checked before anything is stored; duplicates are refused.
- Temporary failures are retried automatically (up to 3 attempts); for anything else, the reason is shown in the app.
- The AI models run locally: no document leaves our machine.

📖 Glossary

API server	The “front desk” that receives and checks every request.
Background worker	A helper that processes jobs from the to-do list.
Passage	A short piece of a document, about half a page.
Meaning fingerprint	Numbers that capture what a passage is about (technically, an “embedding”).
Database	PostgreSQL with pgvector: stores text, fingerprints and the word index in one place.